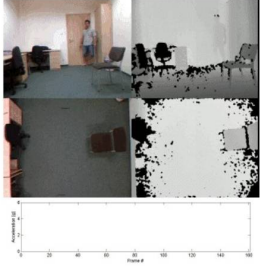


UR Fall Detection Dataset

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This dataset contains 70 (30 falls + 40 activities of daily living) sequences. Fall events are recorded with 2 Microsoft Kinect cameras and corresponding accelerometric data. ADL events are recorded with only one device (camera 0) and accelerometer. Sensor data was collected using PS Move (60Hz) and x-IMU (256Hz) devices. The dataset is organized as follows. Each row contains sequence of depth and RGB images for camera 0 and camera 1 (parallel to the floor and ceiling mounted, respectively), synchronization data, and raw accelerometer data. Each video stream is stored in separate zip archive in form of png image sequence. Depth data is stored in PNG16 format and should be rescaled:

$$d = \frac{C_i P(x, y)}{65535}$$

where d is depth in millimeters, C_i is scale ratio for i -th camera and $P(x, y)$ is pixel value at position (x, y) of PNG16 image. Fall sequences: $C_0 = 6000$ and $C_1 = 3640$. ADLs: $C_0 = 7000$

Synchronization data contains: frame number, time in milliseconds since sequence start and interpolated accelerometric data ($SV_{i \in \{0,1\}}$) - corresponding to image frame. Note that the cameras are recorded independently, so they are not strictly synchronized (synchronization based on nearest timestamp value).

Raw accelerometric data contains time in milliseconds since sequence start and accelerometer data: $SV_{i \in \{0,1\}}, A_{x_i}, A_{y_i}, A_{z_i}$. All accelerometer data are in gravity units (g). Total sum vector is calculated as follows:

$$SV_{i \in \{0,1\}} = \sqrt{A_{x_i}^2 + A_{y_i}^2 + A_{z_i}^2}$$

If you use the dataset, please cite the following work:

Bogdan Kwolek, Michał Kępski, Human fall detection on embedded platform using depth maps and wireless accelerometer, Computer Methods and Programs in Biomedicine, Volume 117, Issue 3, December 2014, Pages 489-501, ISSN 0169-2607 [\[Link\]](#)

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Fall sequences:

#	Depth data		RGB data		Synchronization data	Accelerometer data	Video
	Camera 0	Camera 1	Camera 0	Camera 1			
01	fall-01-cam0-d.zip	fall-01-cam1-d.zip	fall-01-cam0-rgb.zip	fall-01-cam1-rgb.zip	fall-01-data.csv	fall-01-acc.csv	cam0 cam1
02	fall-02-cam0-d.zip	fall-02-cam1-d.zip	fall-02-cam0-rgb.zip	fall-02-cam1-rgb.zip	fall-02-data.csv	fall-02-acc.csv	cam0 cam1
03	fall-03-cam0-d.zip	fall-03-cam1-d.zip	fall-03-cam0-rgb.zip	fall-03-cam1-rgb.zip	fall-03-data.csv	fall-03-acc.csv	cam0 cam1
04	fall-04-cam0-d.zip	fall-04-cam1-d.zip	fall-04-cam0-rgb.zip	fall-04-cam1-rgb.zip	fall-04-data.csv	fall-04-acc.csv	cam0 cam1
05	fall-05-cam0-d.zip	fall-05-cam1-d.zip	fall-05-cam0-rgb.zip	fall-05-cam1-rgb.zip	fall-05-data.csv	fall-05-acc.csv	cam0 cam1
06	fall-06-cam0-d.zip	fall-06-cam1-d.zip	fall-06-cam0-rgb.zip	fall-06-cam1-rgb.zip	fall-06-data.csv	fall-06-acc.csv	cam0 cam1
07	fall-07-cam0-d.zip	fall-07-cam1-d.zip	fall-07-cam0-rgb.zip	fall-07-cam1-rgb.zip	fall-07-data.csv	fall-07-acc.csv	cam0 cam1
08	fall-08-cam0-d.zip	fall-08-cam1-d.zip	fall-08-cam0-rgb.zip	fall-08-cam1-rgb.zip	fall-08-data.csv	fall-08-acc.csv	cam0 cam1
09	fall-09-cam0-d.zip	fall-09-cam1-d.zip	fall-09-cam0-rgb.zip	fall-09-cam1-rgb.zip	fall-09-data.csv	fall-09-acc.csv	cam0 cam1
10	fall-10-cam0-d.zip	fall-10-cam1-d.zip	fall-10-cam0-rgb.zip	fall-10-cam1-rgb.zip	fall-10-data.csv	fall-10-acc.csv	cam0 cam1
11	fall-11-cam0-d.zip	fall-11-cam1-d.zip	fall-11-cam0-rgb.zip	fall-11-cam1-rgb.zip	fall-11-data.csv	fall-11-acc.csv	cam0 cam1
12	fall-12-cam0-d.zip	fall-12-cam1-d.zip	fall-12-cam0-rgb.zip	fall-12-cam1-rgb.zip	fall-12-data.csv	fall-12-acc.csv	cam0 cam1
13	fall-13-cam0-d.zip	fall-13-cam1-d.zip	fall-13-cam0-rgb.zip	fall-13-cam1-rgb.zip	fall-13-data.csv	fall-13-acc.csv	cam0 cam1
14	fall-14-cam0-d.zip	fall-14-cam1-d.zip	fall-14-cam0-rgb.zip	fall-14-cam1-rgb.zip	fall-14-data.csv	fall-14-acc.csv	cam0 cam1
15	fall-15-cam0-d.zip	fall-15-cam1-d.zip	fall-15-cam0-rgb.zip	fall-15-cam1-rgb.zip	fall-15-data.csv	fall-15-acc.csv	cam0 cam1
16	fall-16-cam0-d.zip	fall-16-cam1-d.zip	fall-16-cam0-rgb.zip	fall-16-cam1-rgb.zip	fall-16-data.csv	fall-16-acc.csv	cam0 cam1
17	fall-17-cam0-d.zip	fall-17-cam1-d.zip	fall-17-cam0-rgb.zip	fall-17-cam1-rgb.zip	fall-17-data.csv	fall-17-acc.csv	cam0 cam1
18	fall-18-cam0-d.zip	fall-18-cam1-d.zip	fall-18-cam0-rgb.zip	fall-18-cam1-rgb.zip	fall-18-data.csv	fall-18-acc.csv	cam0 cam1
19	fall-19-cam0-d.zip	fall-19-cam1-d.zip	fall-19-cam0-rgb.zip	fall-19-cam1-rgb.zip	fall-19-data.csv	fall-19-acc.csv	cam0 cam1
20	fall-20-cam0-d.zip	fall-20-cam1-d.zip	fall-20-cam0-rgb.zip	fall-20-cam1-rgb.zip	fall-20-data.csv	fall-20-acc.csv	cam0 cam1
21	fall-21-cam0-d.zip	fall-21-cam1-d.zip	fall-21-cam0-rgb.zip	fall-21-cam1-rgb.zip	fall-21-data.csv	fall-21-acc.csv	cam0 cam1
22	fall-22-cam0-d.zip	fall-22-cam1-d.zip	fall-22-cam0-rgb.zip	fall-22-cam1-rgb.zip	fall-22-data.csv	fall-22-acc.csv	cam0 cam1
23	fall-23-cam0-d.zip	fall-23-cam1-d.zip	fall-23-cam0-rgb.zip	fall-23-cam1-rgb.zip	fall-23-data.csv	fall-23-acc.csv	cam0 cam1
24	fall-24-cam0-d.zip	fall-24-cam1-d.zip	fall-24-cam0-rgb.zip	fall-24-cam1-rgb.zip	fall-24-data.csv	fall-24-acc.csv	cam0 cam1
25	fall-25-cam0-d.zip	fall-25-cam1-d.zip	fall-25-cam0-rgb.zip	fall-25-cam1-rgb.zip	fall-25-data.csv	fall-25-acc.csv	cam0 cam1
26	fall-26-cam0-d.zip	fall-26-cam1-d.zip	fall-26-cam0-rgb.zip	fall-26-cam1-rgb.zip	fall-26-data.csv	fall-26-acc.csv	cam0 cam1
27	fall-27-cam0-d.zip	fall-27-cam1-d.zip	fall-27-cam0-rgb.zip	fall-27-cam1-rgb.zip	fall-27-data.csv	fall-27-acc.csv	cam0 cam1
28	fall-28-cam0-d.zip	fall-28-cam1-d.zip	fall-28-cam0-rgb.zip	fall-28-cam1-rgb.zip	fall-28-data.csv	fall-28-acc.csv	cam0 cam1
29	fall-29-cam0-d.zip	fall-29-cam1-d.zip	fall-29-cam0-rgb.zip	fall-29-cam1-rgb.zip	fall-29-data.csv	fall-29-acc.csv	cam0 cam1
30	fall-30-cam0-d.zip	fall-30-cam1-d.zip	fall-30-cam0-rgb.zip	fall-30-cam1-rgb.zip	fall-30-data.csv	fall-30-acc.csv	cam0 cam1

Activities of Daily Living (ADL) sequences:

#	Depth data		RGB data		Synchronization data	Accelerometer data	Video
	Camera 0	(not available)	Camera 0	(not available)			
01	adl-01-cam0-d.zip	*	adl-01-cam0-rgb.zip	*	adl-01-data.csv	adl-01-acc.csv	cam0
02	adl-02-cam0-d.zip	*	adl-02-cam0-rgb.zip	*	adl-02-data.csv	adl-02-acc.csv	cam0
03	adl-03-cam0-d.zip	*	adl-03-cam0-rgb.zip	*	adl-03-data.csv	adl-03-acc.csv	cam0
04	adl-04-cam0-d.zip	*	adl-04-cam0-rgb.zip	*	adl-04-data.csv	adl-04-acc.csv	cam0
05	adl-05-cam0-d.zip	*	adl-05-cam0-rgb.zip	*	adl-05-data.csv	adl-05-acc.csv	cam0
06	adl-06-cam0-d.zip	*	adl-06-cam0-rgb.zip	*	adl-06-data.csv	adl-06-acc.csv	cam0
07	adl-07-cam0-d.zip	*	adl-07-cam0-rgb.zip	*	adl-07-data.csv	adl-07-acc.csv	cam0
08	adl-08-cam0-d.zip	*	adl-08-cam0-rgb.zip	*	adl-08-data.csv	adl-08-acc.csv	cam0
09	adl-09-cam0-d.zip	*	adl-09-cam0-rgb.zip	*	adl-09-data.csv	adl-09-acc.csv	cam0
10	adl-10-cam0-d.zip	*	adl-10-cam0-rgb.zip	*	adl-10-data.csv	adl-10-acc.csv	cam0
11	adl-11-cam0-d.zip	*	adl-11-cam0-rgb.zip	*	adl-11-data.csv	adl-11-acc.csv	cam0
12	adl-12-cam0-d.zip	*	adl-12-cam0-rgb.zip	*	adl-12-data.csv	adl-12-acc.csv	cam0
13	adl-13-cam0-d.zip	*	adl-13-cam0-rgb.zip	*	adl-13-data.csv	adl-13-acc.csv	cam0
14	adl-14-cam0-d.zip	*	adl-14-cam0-rgb.zip	*	adl-14-data.csv	adl-14-acc.csv	cam0
15	adl-15-cam0-d.zip	*	adl-15-cam0-rgb.zip	*	adl-15-data.csv	adl-15-acc.csv	cam0
16	adl-16-cam0-d.zip	*	adl-16-cam0-rgb.zip	*	adl-16-data.csv	adl-16-acc.csv	cam0
17	adl-17-cam0-d.zip	*	adl-17-cam0-rgb.zip	*	adl-17-data.csv	adl-17-acc.csv	cam0
18	adl-18-cam0-d.zip	*	adl-18-cam0-rgb.zip	*	adl-18-data.csv	adl-18-acc.csv	cam0
19	adl-19-cam0-d.zip	*	adl-19-cam0-rgb.zip	*	adl-19-data.csv	adl-19-acc.csv	cam0
20	adl-20-cam0-d.zip	*	adl-20-cam0-rgb.zip	*	adl-20-data.csv	adl-20-acc.csv	cam0
21	adl-21-cam0-d.zip	*	adl-21-cam0-rgb.zip	*	adl-21-data.csv	adl-21-acc.csv	cam0
22	adl-22-cam0-d.zip	*	adl-22-cam0-rgb.zip	*	adl-22-data.csv	adl-22-acc.csv	cam0
23	adl-23-cam0-d.zip	*	adl-23-cam0-rgb.zip	*	adl-23-data.csv	adl-23-acc.csv	cam0
24	adl-24-cam0-d.zip	*	adl-24-cam0-rgb.zip	*	adl-24-data.csv	adl-24-acc.csv	cam0
25	adl-25-cam0-d.zip	*	adl-25-cam0-rgb.zip	*	adl-25-data.csv	adl-25-acc.csv	cam0
26	adl-26-cam0-d.zip	*	adl-26-cam0-rgb.zip	*	adl-26-data.csv	adl-26-acc.csv	cam0
27	adl-27-cam0-d.zip	*	adl-27-cam0-rgb.zip	*	adl-27-data.csv	adl-27-acc.csv	cam0
28	adl-28-cam0-d.zip	*	adl-28-cam0-rgb.zip	*	adl-28-data.csv	adl-28-acc.csv	cam0
29	adl-29-cam0-d.zip	*	adl-29-cam0-rgb.zip	*	adl-29-data.csv	adl-29-acc.csv	cam0
30	adl-30-cam0-d.zip	*	adl-30-cam0-rgb.zip	*	adl-30-data.csv	adl-30-acc.csv	cam0
31	adl-31-cam0-d.zip	*	adl-31-cam0-rgb.zip	*	adl-31-data.csv	adl-31-acc.csv	cam0
32	adl-32-cam0-d.zip	*	adl-32-cam0-rgb.zip	*	adl-32-data.csv	adl-32-acc.csv	cam0
33	adl-33-cam0-d.zip	*	adl-33-cam0-rgb.zip	*	adl-33-data.csv	adl-33-acc.csv	cam0
34	adl-34-cam0-d.zip	*	adl-34-cam0-rgb.zip	*	adl-34-data.csv	adl-34-acc.csv	cam0
35	adl-35-cam0-d.zip	*	adl-35-cam0-rgb.zip	*	adl-35-data.csv	adl-35-acc.csv	cam0
36	adl-36-cam0-d.zip	*	adl-36-cam0-rgb.zip	*	adl-36-data.csv	adl-36-acc.csv	cam0
37	adl-37-cam0-d.zip	*	adl-37-cam0-rgb.zip	*	adl-37-data.csv	adl-37-acc.csv	cam0
38	adl-38-cam0-d.zip	*	adl-38-cam0-rgb.zip	*	adl-38-data.csv	adl-38-acc.csv	cam0
39	adl-39-cam0-d.zip	*	adl-39-cam0-rgb.zip	*	adl-39-data.csv	adl-39-acc.csv	cam0
40	adl-40-cam0-d.zip	*	adl-40-cam0-rgb.zip	*	adl-40-data.csv	adl-40-acc.csv	cam0

Extracted features:

urfall-cam0-falls.csv	urfall-cam0-adls.csv
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Extracted features from depth maps are stored in CSV format. Each row contains one sample of data corresponding to one depth image. The columns from left to right are organized as follows:

- sequence name - camera name is omitted, because all of the samples are from the front camera ('fall-01-cam0-d' is 'fall-01', 'adl-01-cam0-d' is 'adl-01' and so on),
- frame number - corresponding to number in sequence,
- label - describes human posture in the depth frame; '-1' means person is not lying, '1' means person is lying on the ground; '0' is temporary pose, when person "is falling", we don't use '0' frames in classification,
- HeightWidthRatio - bounding box height to width ratio,
- MajorMinorRatio - major to minor axis ratio, computed from BLOB of segmented person,
- BoundingBoxOccupancy - ratio of how bounding box is occupied by person's pixels,
- MaxStdXZ - standard deviation of pixels from the centroid for the abscissa (X axis) and the depth (Z axis), respectively (it is computed on segmented person transformed to the 3D point cloud),
- HHmaxRatio - human height in frame to human height while standing ratio,
- H - actual height (in mm),
- D - distance of person center to the floor (in mm),
- P40 - ratio of the number of the point clouds belonging to the cuboid of 40 cm height and placed on the floor to the number of the point clouds belonging to the cuboid of height equal to person's height.